Priyanka Asthana

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Portfolio LinkedIn GitHub

Research Interests

Intersection of ML, combinatorial optimization, and NLP —Specifically, how constrained reasoning and algorithmic decision-making can be embedded into systems for real-world deployment in healthcare.

Education

Dr. A.P.J. Abdul Kalam Technical University (AKTU) **2022-2026**

Institute: Ashoka Institute of Technology and Management, Varanasi

BTech (Hons.) Computer Science and Engineering | Minor: Robotics | CGPA: 9.01/10

- **Relevant Coursework:** Algorithms, Data Structures, Discrete Mathematics, Theory of Computation, Automata Theory, OS, Compiler Design (Basics), Linear Algebra, Probability & Statistics, Optimization Methods.

Publications

Priyanka, A., Harshita, K., & Shanker Maurya, L. (2026). Priority-Weighted Spatial Framework for Emergency Blood Allocation. Zenodo. https://doi.org/10.5281/zenodo.18149564

- Developed novel Spatial–Temporal GeoAI framework combining priority-weighted geographic scoring with time-bound token reservation mechanism for emergency blood allocation.
- Manuscript under review at Springer Nature (*Journal of Computational Social Science*).

Patents & Intellectual Property

Patent Application Filed — *AI-Driven Conversational Medical Voice Assistant*

Indian Patent Application No. - 202611012869, filed Feb 2026

- Designed an NLP-based voice interaction system integrating speech recognition, semantic parsing, and constrained AI reasoning for real-time symptom assessment and structured report generation.

Leadership

Student Council President | AITM Varanasi **05/2025–05/2026**

- Represented AITM campus students; coordinated academic and institutional initiatives.

Patriotic Youth Ambassador | Veterans India **08/2025–09/2025**

- Led youth campaigns promoting veteran welfare, engaging 200+ participants.

Projects

Kashivani: Multi-Agent System for Cultural Knowledge Generation ↗ **09/2025 – 10/2025**

- Developed a multi-agent NLP system for culturally grounded knowledge generation focused on Varanasi's heritage.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline to reduce hallucinations by 40%.

DiagnoseAI: NLP DRIVEN MEDICAL SYMPTOM ASSESSMENT SYSTEM ↗ **04/2025 – 11/2025**

- Designed a voice-first NLP system for real-time symptom assessment using speech-to-text + constrained LLM pipeline for real-time symptom.
- Processed 500+ patient scenarios with 92% accuracy.

Tokenized GeoAI Blood Allocation Framework (RAKT) ↗ **12/2024 – 11/2025**

- Engineered Spatial-Temporal GeoAI system with priority-weighted GPS clustering + time-bound token reservations for emergency blood banks.
- Achieved 100% high-priority allocation and 6.6% distance reduction in scarcity simulations.

Experience

IIT BHU| AI Intern (NLP Engineer) **02/2026 – 04/2026**

- Engineered NLP-driven test automation, processing 1,000+ text inputs for semantic accuracy.
- Detected and resolved 20+ NLP parsing issues in real-time systems, boosting reliability by 25%.

Bharat Space Education Research Centre | AI Research Intern **12/2025 – 01/2026**

- Developed AI-driven control algorithms for UAS across 5 experimental modules, optimizing trajectory prediction accuracy by 15%.
- Applied ML to real-time sensor data fusion, reducing subsystem latency by 20% in simulated space missions.

Hindalco Industries (Aditya Birla Group) | IT Intern **07/2025 – 08/2025**

- Engineered analytical dashboards and automated ERP data pipelines, accelerating API/query performance by 10% for industrial operations.
- Optimized data flows across enterprise systems and reducing manual processing time by 25%.

Skills & abilities

- Algorithms & Theory**:** Optimization basics, graph algorithms, combinatorics, complexity analysis.
- AI/ML: Model training, feature engineering, constrained LLM reasoning, NLP pipelines.
- Frameworks & Libraries**:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV.
- Programming: Java (DSA, OOP), Python (ML, Data Processing), JavaScript.

Certifications

- Microsoft Azure AI-102 (Designing and Implementing AI Solutions): Microsoft Industrial Training
- Microsoft Azure AI-900 (AI Fundamentals): Microsoft & Infosys Training Program
- Oracle AI Vector Search Professional Certification (Applied AI Systems, 2025)
- McKinsey Forward Program (2025)